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Health-seeking challenges among older adults with hypertension in Nepal: a phenomenological study

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Abstract

Background Hypertension is a leading non-communicable disease in Nepal, with a rising burden among older adults and persistently low rates of blood pressure control. While several studies have explored healthcare-seeking behavior in the general population, little is known about the specific challenges faced by older adults with hypertension. In particular, the perspectives of older adults with hypertension, family caregivers and healthcare providers-key actors in the care process are rarely explored. Therefore, this study aims to explore challenges to healthcare-seeking among older adults with hypertension in Nepal from the perspective of older adults, family caregivers, and healthcare providers.

Method We adopted a qualitative study design to collect the required information. Trained researchers conducted 24 In Depth Interviews (IDIs) with older adults, healthcare providers, and facilitated 2 Focus Group Discussions (FGDs) with 18 caregivers. Interviews and FGDs were held in private settings using a semi-structured guideline based on the socio-ecological model. The interviews and FGDs were conducted in the Nepali language. The audio-recorded sessions were then transcribed verbatim and subsequently translated into English before analysis, to then apply both inductive and deductive coding. We then analyzed the data using thematic analysis with Taguette software.

Result Applying the Socio-Ecological Model, we explored multi-level challenges to seeking care as perceived by older adults with hypertension, family care givers and healthcare providers in Central Nepal. At the individual level, perceived physical and cognitive decline, low health autonomy, and competing priorities as limiting their ability to seek care, while healthcare providers viewed these same factors as major challenges to timely service utilization. Interpersonal challenges were perceived in terms of limited family support and poor communication between older adults with hypertension and health care providers. At the community level, inadequate infrastructure and transportation difficulties restrict access. Organizational challenges were perceived as insufficient geriatric services, fragmented care pathways, and shortages of health staff, with providers particularly emphasizing these system constraints. Finally, at the policy level, perceived exclusion from health insurance, lack of structured programs, and inefficiencies in medication supply and enforcement as significant challenges.

Conclusion To improve hypertension management, a coordinated response is needed by investing in geriatric-friendly services, empowering older adults with hypertension, strengthening family and community support, and

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closing the policy-to-practice gap. As Nepal's population ages, addressing these issues is vital to achieving equitable, age-inclusive healthcare.

Keywords Hypertension, Blood pressure control, Older adults, Medication adherence, BMI, Socio-ecological model, Care-seeking behavior, Challenges to healthcare

Background

Hypertension is one of the most prevalent Non-Communicable Diseases (NCDs) globally and a leading risk factor for cardiovascular morbidity and mortality [1]. In Nepal, the burden of hypertension is rising, particularly among older adults [2]. According to the Nepal Demographic and Health Survey (NDHS) 2022, about 46% of women and 42% of men aged 60 and above are hypertensive [2]. This age group represents a growing portion of the population due to demographic transitions, underscoring the urgent need to address hypertension management in this vulnerable group [3].

Although hypertension can be effectively managed with lifestyle changes and medications, blood pressure (BP) control in Nepal remains low [4]. Only around 35–38% of those receiving treatment have their BP under control [5]. Similarly, the Dhulikhel Heart Study found that only 35.3% of hypertensive patients on treatment had controlled BP [6]. For older populations who often face multiple comorbidities and increased healthcare needs, these figures point to critical gaps not only in maintaining their health status but also in accessing care [7].

Nepal's health care system is challenged by rising NCDs burden amid limited resources, infrastructure and geriatric focused services [8, 9]. Older adults face unique challenges such as financial constraints, lack of transportation, low health literacy, and limited access to age-friendly health services [10, 11]. While national surveys and hospital-based studies offer estimates of hypertension prevalence, they provide limited insight into age-specific blood pressure control and the unique challenges older adults face in accessing and maintaining care.

Moreover, research in Nepal has primarily focused on the general adult population, often neglecting the specific experiences and vulnerabilities of older adults. Although some studies have explored healthcare challenges among older adults more broadly, few have focused specifically on those living with hypertension [10]. Furthermore, existing evidence rarely includes perspectives from family caregivers and frontline healthcare providers, whose voices are crucial to understanding the full scope of the issue.

This study aimed to explore the health-seeking challenges in blood pressure control faced by older adults with hypertension in Nepal. By including perspectives from older adults, family caregivers, and healthcare providers within a specific municipality, the study seeks to provide clear evidence to improve hypertension management and ensure continuous care for this vulnerable and growing population.

Method

Study design

We used a phenomenological qualitative study design to explore the challenges related to seeking care for hypertension as it enables us to explore individual's lived experiences, emotions, and challenges [12]. The study was reported following the Standards for Reporting Qualitative Research (SRQR) guidelines, and a completed checklist is provided in the supplementary file 1. The socio-ecological model (SEM) served as the guiding framework to examine how individual, interpersonal, community, and systemic factors influence health-seeking behavior. The SEM provides a comprehensive framework to capture the multi-level influences on health-seeking behavior, enabling a deeper understanding of the challenges experienced by this population.

Study site

We conducted the study in Dhulikhel Municipality, Kavrepalanchowk district, central Nepal, about 30 km east of Kathmandu, the capital city [13]. The municipality spans 55 square kilometers and houses 34,858 people, of whom 3,689 are older adults aged 60 years and above. Its population is ethnically diverse and represents a wide range of socioeconomic backgrounds. We purposely selected Dhulikhel municipality because it has a sizable and varied older adult population, which provides an appropriate context to explore differences in hypertension management. The municipality also offered logistical feasibility for fieldwork, making it a strategic and practical choice. These factors together strengthen the relevance of our study findings not only for Dhulikhel municipality but also for other semi-urban municipalities in Nepal.

Study participants and sample size

We employed purposive sampling methods to recruit participants who capture diverse perspectives on challenges to seek hypertension care. We included older adults who met the definition of senior citizens according to the Senior Citizens Act, 2063 of Nepal (≥ 60 years), reported having hypertension in the Healthy City Initiative dataset, and had a measured blood pressure ≥ 140/90 mmHg at the time of recruitment [14]. We included caregivers if they were family members or relatives who

directly provided care to these older adults. We selected health care providers purposefully based on who is providing health care services to hypertensive patients in community health care settings, local health care centers and tertiary hospitals. Health care workers included clinical pharmacist (n = 1), pharmacovigilance officer (n = 1), FCHVs (n = 2), Health post incharge (n = 2), Nursing incharge (n = 2), senior Auxiliary Nurse Midwife (ANM, n = 1), Health Assistant (n = 2), Home based care enumerator (n = 1), Medical officer (n = 1), Municipality health coordinator- (n = 1), Consultant physician (n = 1), DM cardiologist residence (n = 1), and physiotherapy officer (n = 1). FCHVs are a group of community-based health workers in Nepal who play a vital role in delivering health services at local level [15]. In total, 42 participants were recruited for this study.

The sample size was guided by the principles of code and meaning saturation to the point at which no new themes or insights emerge from the data [16]. Data saturation was considered achieved when repetition of codes occurred, no new codes emerged from additional interviews, and sufficient depth was reached to capture diverse experiences and perceptions of participants. We ensured variation across demographic and contextual factors so that a wide range of perspectives was represented. Saturation was carried out through regular team discussions, comparison of emerging themes, and cross-checking of coded data, which confirmed that further data collection was unlikely to generate new insights.

Participant recruitment and sampling strategy

We used the Healthy City Initiative Dataset of Dhulikhel Municipality to identify eligible participants [17]. We purposefully selected participants and healthcare providers to capture diverse perspectives on challenges in seeking hypertension care. Hospital staff helped us identify healthcare providers who were providing health care services to hypertensive patients as well as older adults, and we contacted patients by phone to schedule convenient interview times. For caregivers, we coordinated with FCHVs to arrange suitable interviews. We conducted interviews with healthcare providers at their workplaces and with patients and caregivers near their households. We contacted all participants in advance, and none declined to participate.

Data collection tools and techniques

We conducted FGDs and IDIs, using a semi-structured interview guide. We developed the interview guidelines based on key constructs from the SEM and refined them through multiple team discussions to ensure relevance and comprehensiveness (Supplementary Material 3). We then conducted a pilot test with a small group of participants to assess the understandability, flow, and feasibility of the questions, and we incorporated their feedback to improve the final version of the guidelines. We did not include data from the pilot in the main analysis. Between September 2024 and March 2025, three trained qualitative researchers (PS, YA, and IP), all native Nepali speakers with prior experience in qualitative data collection, conducted the interviews. All interviews were held in Nepali, in private and quiet settings to ensure participant comfort and confidentiality. Each session lasted between 30 and 60 min. We provided a small food package to older adults and their caregivers as a token of appreciation for their time and valuable participation in the interviews.

Data management and analysis

All audio-recorded interviews were transcribed into Nepali verbatim and subsequently translated into English before analysis. Transcripts and translations were reviewed for accuracy by cross-checking with the original audio recordings. To ensure participant anonymity, each individual was assigned an alphanumeric code, and transcripts were de-identified prior to analysis.

We employed thematic analysis for qualitative data, a systematic method of organizing and interpreting data within a theme. The analysis began with familiarization, during which the research team read and re-read the transcripts to develop an initial understanding of key themes. Initial codes were then generated to capture salient ideas and relevant content. Both inductive codes, emerging directly from the data, and deductive codes, based on the interview guide and the socio-ecological model, were included in the codebook, which was iteratively refined through frequent team discussions. Ultimately, the analysis yielded five themes and twenty codes.

The Socio-Ecological Model, which recognizes that health behaviors and experiences are influenced by multiple, interacting levels of factors such as individual, interpersonal, community, organizational, and policy, served as the analytical framework for organizing and interpreting the data. By integrating the SEM into our analysis, we were able to examine how challenges in hypertension care-seeking operated within and across these levels [18]. Thematic analysis was conducted within the structure of the Socio-Ecological Model, organizing data across five levels: individual, interpersonal, community, organizational, and policy (Fig. 1). Similar codes were grouped into and were further organized into overarching themes (Supplementary Material 2). A matrix was developed to map the data according to these domains, ensuring consistency and alignment with the theoretical framework. Qualitative data were managed and analyzed using Taguette, an open-source qualitative data analysis software.

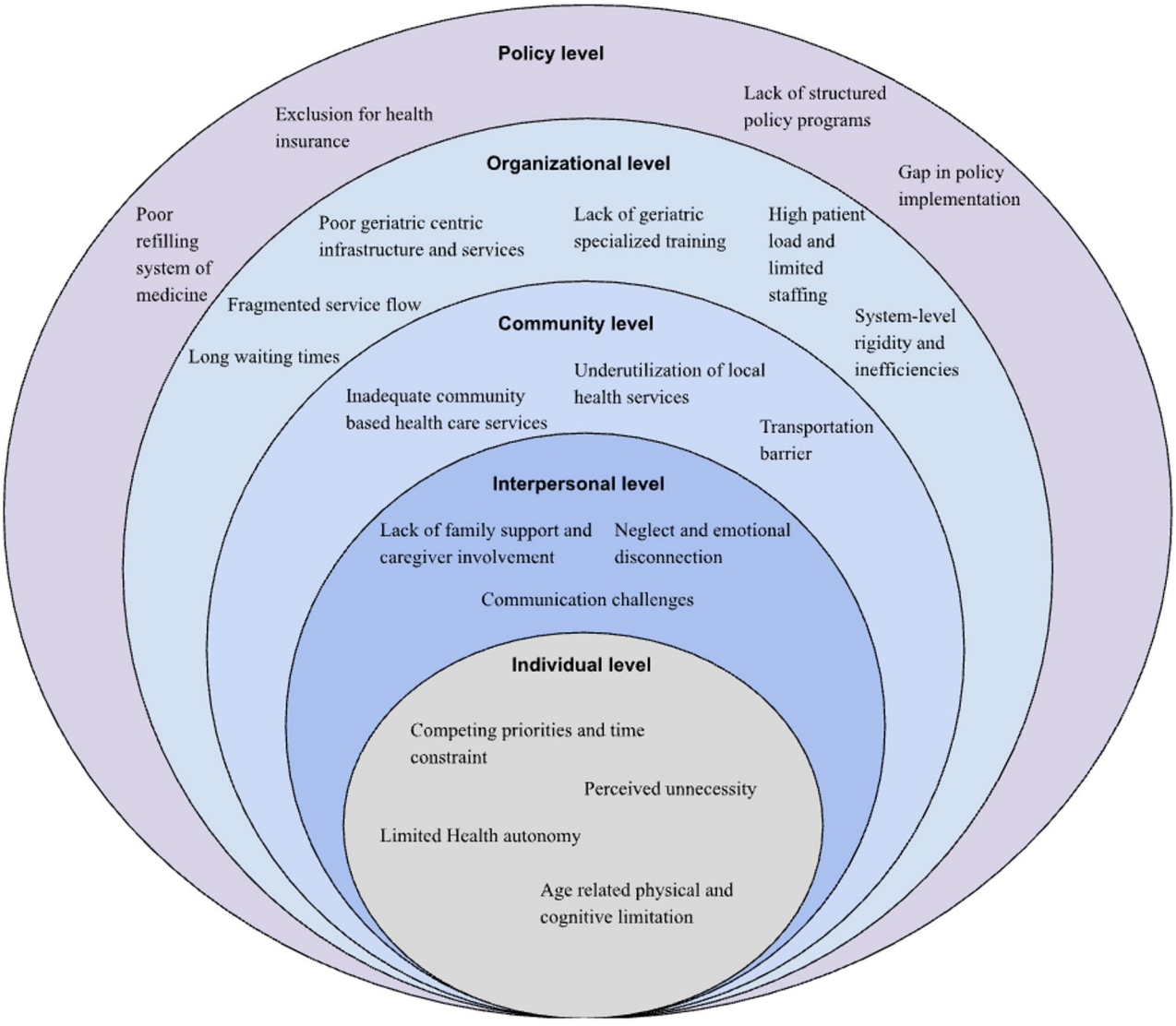


Fig. 1 Socio-ecological model to explore health-seeking challenges among older adults with hypertension

Trustworthiness of the study

Our research team included three members. PS, the first author, led the qualitative research process. PS, YA, and IP conducted the interviews, while PS and YA, both trained public health professionals with prior qualitative experience, also developed and refined the interview guides, coding, and theme generation. We ensured objectivity by maintaining no personal or professional relationships with study participants. Experts in public health, geriatric health, and qualitative research reviewed the interview guide, which strengthened its content validity and clarity. We applied iterative questioning, audio-recorded all interviews, and reached coding decisions through team consensus. To enhance reflexivity, PS and YA wrote memos and maintained research journals after each interview and throughout the analysis phase, documenting initial impressions, emergent ideas, and potential biases. These practices enabled us to critically examine data interpretation. PS, YA, and IP met regularly to discuss emerging themes and refine interpretations collectively.

Result

Among the seven older adults with hypertension, the mean age was 72.1 (4.7) years (Table 1). More than half of the older hypertensive patients (57.2%) were female. 42% of the older hypertensive patients had received secondary education or more and more than half of the participants were of janjati ethnicity. Among health care workers, the mean age was 38.7 (8.7) years. 70% of the health care workers were female and the mean years of experience was 10.2 (9.3) years.In two of the FGDs, the mean age of the caretaker was 42.6 (13.1) years. 77% of

Table 1 Socio demographic characteristics of the participants

Characteristics	Older adults with Hypertension (*n* = 7) n(%)	Caretakers (*n* = 18)
Age (in years) Mean (SD)	72.1 (4.7)	42.6 (13.07)
Sex		
Male	3 (42.8)	4 (22.2)
Female	4 (57.2)	14 (77.8)
Educational status		
Not able to read and write	1 (14.2)	2 (11.1)
Informal education	1 (14.2)	-
Basic education	2 (28.5)	4 (22.2)
Secondary education and above	3 (42.8)	12 (66.7)
Ethnicity		
Bhrahmin/Chettri	4 (57.1)	18 (100)
Janjati	3 (42.9)	-
Health care provider(*n* = 17)		
Age (in years) Mean (SD)	38.7 (8.7)	
Sex		
Male	5 (29.4)	
Female	12 (70.6)	
Years of experience Mean (SD)	10.2 (9.3)	

the participants were female. About 67% of the participants had secondary education and above.

Our study explored that the health-seeking among older adults with hypertension is shaped by the dynamic interplay between individual capacities, family support, and systemic structures. While a patient's personal beliefs, competing priorities, and physical limitations directly constrained their engagement with health seeking, these individual-level barriers were often magnified by the absence of caregiver support. Caregiver's capacity to provide care was influenced by family dynamics, migration of younger members, and emotional involvement, highlighting how interpersonal contexts mediate a patient's health behaviors. Healthcare providers emphasized that even when patients and caregivers were motivated, organizational inefficiencies, fragmented services, and policy gaps frequently disrupted continuity of care. These intersecting perspectives underscore that challenges are not isolated but mutually reinforcing across individual, interpersonal, community, organizational, and policy levels, shaping challenges for health seeking hypertension among this population.

Individual level challenges to health-seeking

Several challenges emerged at the individual level that negatively influenced participants' involvement in seeking care. These challenges reflect personal beliefs, knowledge, motivations, and physical limitations that shape health-seeking behavior and disease management.

Competing priorities and time constraint

Individual obligations, such as caregiving duties and household roles, limited participants' ability to prioritize their own health needs. Particularly among elderly participants, dependency on family members for accompaniment and care support further restricted their access to timely care.

> *My son goes to work, and my husband has to go out. My daughter-in-law takes care of our young granddaughter because there's no one else at home. There are only four of us in the family. When will I even get the time? [p4, patient,64 years]*

Health care providers also explained about these challenges noting that older adults often deprioritize their own health not only in terms of seeking care but also doing preventable work in favor of work or family obligations.

> *People should visit the health center regularly for checkups. But often, they give more importance to household work than their health. Some are busy with farming or other jobs, so they may not be able to come on their own. [HCP_13, Medical officer, 33 years]*

Perceived unnecessity of seeking care

Some participants expressed a belief that regular monitoring or treatment was unnecessary unless they experienced acute symptoms. This perception led to delayed or reactive health-seeking behavior, undermining long-term management of chronic conditions such as hypertension.

> *It wasn't that necessary, that's why I didn't go to check my blood pressure. Only when my blood pressure is troubling me, I will go. [p7,patient, 70 years]*
> *They don't have enough information. They only check their blood pressure when something serious happens. They don't realize the importance of regular check-ups. [HCP_09, Health Assistant, 42 years]*

Others held fatalistic attitudes toward their health, viewing hypertension management as futile due to old age or limited life expectancy. This belief discouraged continued adherence to medications or regular health checkups. A healthcare provider noted this sentiment, quoting a patient who said:

> *I am already going to die anyway, why should I continue taking medicine? How long will I even live?*

There is no benefit in taking the medicine. Why should I waste my money? [HCP_09, Health Assistant, 42 years]

Limited health autonomy

Older women in particular lacked autonomy in making decisions about their health care, needing permission or approval from male family members before seeking medical help.

I have to ask my husband or my sons before going to the hospital. Sometimes they say it's not necessary then I stay at home only. [p6, patient, 74 years]

Age related physical and cognitive limitation

Physical ailments such as joint pain, visual or hearing impairment, and general fatigue hindered many older adults from visiting health facilities or maintaining treatment regimens.

After the age of 60, most people have osteoarthritis…because of that, they say their knees hurt and they can't come. Even if they are regular or normal patients, it can still be difficult for them to come for follow-ups. [HCP_04, Nursing In-charge, 43 years]

It's hard to make them understand. Hearing loss is a big issue among old age. Even when we use gestures or nod while explaining, many patients still don't get it. For example, if we tell them to take two tablets of amlodipine, they may still take only one. So, communication becomes a problem. [HCP_13, Medical Officer, 33 years]

Interpersonal level challenges to health-seeking

The interpersonal environment, particularly the role of caregivers and family members, significantly influenced hypertensive patient's ability and willingness to seek timely and appropriate care. Multiple dimensions of caregiver support emerged, ranging from inconsistent assistance with healthcare navigation to communication challenges and emotional neglect.

Lack of family support and caregiver involvement

The physical presence of caregivers during hospital visits was critical not just for logistical support but also for understanding medical instructions. Patients who visited alone often had difficulty comprehending doctors' advice and were unable to convey accurate information to family members.

If he goes alone, he might not fully understand what the doctor says. When he comes home and we ask what the doctor said, even if the doctor says something bad, he'll say it was good. [C1, caretaker, 38 years]

Increased migration of younger family members and family structure has left many older adults without essential support, making the issue worse.

For senior citizens, this is a big problem. Many young family members have moved away, and those who still live at home often do not give much attention to them. Because everyone is busy with their own work, it has become harder for families to take care of their older members. [HCP_7, Health coordinator, 50 years]

People who depend on their spouse manage fine while their spouse is alive. But after the spouse passes away, life gets very hard. They have no one to go with them to the hospital or care for them at home. It's even harder when their children don't visit. Those in joint families usually eat better and take their medicines on time. But for people living alone, it's really tough. [HCP_01, Nursing In-charge, 27 years]

Neglect and emotional disconnection

Some patients expressed a lack of emotional support and involvement from family members, which contributed to feelings of isolation and hesitancy in seeking care. The absence of encouragement or guidance from family led to delays in decision-making around hospital visits and treatment.

I am unable to decide when to visit the hospital. There is no one to give suggestions or anyone I can ask regarding my health status. [p7, patient,70 years]

The people at home, not even my son, care. My daughter-in-law doesn't care much either. When I'm sick, no one cares. My wife says, 'Go to the hospital, go to the hospital.' I just reply saying 'I' will die here, I won't go to the hospital.' [p7, patient, 70 years].

Communication challenges with health care provider

Communication between patients and healthcare providers was difficult due to age-related hearing loss and patients' reluctance to ask clarifying questions. This resulted in misunderstandings and improper medication use.

"Now, there is a problem in making them understand. Hearing is also a big issue… Sometimes, we say to increase the dose of amlodipine to two tablets, but they still take only one." [HCP_13, Medical officer, 33 years].

"There is the problem of hesitation; even if they don't understand, they don't say 'I didn't understand,' they don't re-ask what it means. They just nod their heads like they understood saying 'yes, yes' and sit quietly." [HCP_09, Health Assistant, 42 years].

Community level challenges in health seeking

Community level factors, including the quality of social relationships, accessibility of local healthcare services, and transportation infrastructure, played a significant role in shaping health-seeking behaviors. A lack of supportive networks and structural limitations constrained patients' ability to manage their conditions effectively.

Inadequate community based health care services

Participants highlighted lack of localized health education efforts and suggested a need for more active engagement from health workers at the community level. They called for health education sessions and regular checkups organized in familiar settings by local health staff.

No one really listens to our family members. We don't need hospital staff to spread awareness—someone from the local health post could gather the elderly at a place like the chautari and share useful information. If information comes from community members, even the elderly will take it more seriously. [p5, caretaker, 35 years]

There is a need for many more home visit programs for measuring blood pressure and diabetes. There are now but there should be much more. [p1, patient, 77 years]

Transportation barrier

Long travel times and inconvenient public transport options posed major challenges for patients requiring regular care. Although insured and motivated to seek care, some were hindered by the difficulty of accessing distant health facilities, especially in the absence of personal transport.

Many people travel from here, and the hospital is a bit far. Since we all have insurance and it's the nearest and best option, if a vehicle could come once a month to take everyone for checkups, it would make things much easier. We have to change 2 buses that come every hour. [p6, patient, 74 years]

From here it takes around 1 h to reach the hospital by public bus. On my own bike, it is much easier. The bus won't come whenever we need it. It comes at 1 pm and another at 4 pm. [p5, patient, 72 years]

Underutilization of local health services

There was a shared perception among participants that health services at local level were underdeveloped or underutilized. Participants suggested that community people should utilize basic care at the community level to reduce unnecessary travel and hospital congestion.

If basic treatment is possible here, then it should be done here. Only when it's not possible, they should refer us elsewhere that way, the crowd there would also decrease. Also, people think everything is only available in big hospitals, so even those who don't need to go there end up going, which adds to the problem. [C10, caretaker, 68 years]

Organization level challenges in health seeking

Health system level challenges, especially those related to service delivery processes, facility infrastructure, and inflexible administrative policies posed significant challenges to timely and dignified care for older adults. These organizational inefficiencies not only prolonged the care seeking process but also discouraged consistent engagement with health services.

Long waiting times

Participants frequently expressed frustration over excessive waiting times throughout the care process, which were especially burdensome for elderly and chronically ill patients. Multiple queues for different service points such as ticketing, consultation, diagnostics, and pharmacy which exhausted patients and discouraged them from seeking routine care.

Everywhere, there are long lines. Just to get one medicine, you have to stand in line multiple times. First, stand in line to get a ticket. Then, stand in another line to get the prescription written. Finally, stand in yet another line to collect the medicine. For just one blood pressure medicine, you have to wait for hours in three different places. Only those who are sick truly understand how frustrating this is. [p6, patient, 74 years]

There's always a line no matter when you go from tickets to the doctor, medicine, and blood tests. Everywhere, you have to wait in line, it takes the whole day from morning to evening. [C7, caretaker, 54 years]

Fragmented service flow

The care process was described as highly fragmented and intensive, requiring patients or their families to simultaneously queue in different service areas. This chaotic structure was especially difficult to navigate for

those without strong family support or sufficient health literacy.

> *These days, when going to the hospital for treatment, four people are needed. One waits at the registration desk, another at the doctor's token line, another goes to the X-ray or lab line based on what needs to be checked. If your family has enough people, it helps. After one person leaves their place, 20 others might already be behind them. You have to ask others to hold your place, or you might lose your turn and even get into arguments. Even when you arrive early, it's still crowded. [C10, caretaker, 68 years]*

Poor geriatric centric infrastructure and services

Although elderly patients are meant to be prioritized, significant implementation gaps remain. Older adults continue to face the same long waits and system inefficiencies as the general population, showing a lack of age-sensitive adaptations within hospital operations.

> *And there's no exception for senior citizens; they also have to wait in line. This issue still persists. Senior citizens often end up being patients, and even though they are considered priority, the hospital still has them wait in line. [C10, caretaker, 68 years]*
> *"Senior citizens belong to a different area, so it is hard to provide them with proper care. Even when senior citizens visit the hospital, it is difficult to find a geriatric-friendly doctor. In our health posts, we haven't been able to provide specialized training or skills to care for the elderly." [HCP_11, Health Coordinator, 50 years].*

Providing separate services exclusively for geriatric patients, such as dedicated geriatric OPDs, can substantially reduce waiting times and improve service efficiency. This segregation allows elderly patients to receive faster care without the fatigue and frustration associated with lengthy queues. As a health care provider explained,

> *As much as possible, they should be provided with separate services like the geriatric OPDs that are currently in use. These should be specifically for geriatric patients only, because if it's just for them, they can receive faster services without having to wait in long queues. When that happens, they are more likely to come; it gets easier. They just have to come and go, no need to spend a lot of time. If they can get it done quickly, they will manage to come. The main problem is when they have to wait, that's what they find difficult. [IDI12, Home based care enumerator, 30 years]]*
>
> *Patients should be seen by MDGP physicians and residents, with MDGPs acting as consultants and being present in the area. Cardiac cases should be referred to cardiology, gastrointestinal issues to endoscopy, and fever cases to medicine specialists. The geriatric OPD should cover all these services. There should also be a designated area for vision checks, including astigmatism, with an ophthalmologist available. A dedicated ophthalmologist is essential for the geriatric population. Additionally, a patient-friendly environment must be ensured. [HCP_1, Nursing In-charge, 27 years]*

However, beyond service organization, the hospital infrastructure itself poses challenges for elderly patients. Even facilities housed in one-story buildings lack necessary geriatric-friendly features. Navigating stairs without proper support or access to wheelchairs and stretchers creates significant physical challenges.

> *Even though it's a one-story building, it's not geriatric friendly. You have to go up and down stairs, and there is no proper arrangement for a wheelchair or stretcher. The building is 31 years old, so it needs proper renovation for elderly people. Yes, inside our facility, there should be arrangements like wheelchairs to help patients go to the washroom independently. Now is the time to develop new concepts and plans for these kinds of needs. [HCP_13, Health Coordinator, 50 years]*

Lack of geriatric specialized training

There was a consensus that geriatric patients require specialized care, but current staff and facilities lacked the training and resources to provide this adequately. Many health posts and hospitals did not have geriatric specialists or staff trained in elderly care, reducing the quality of care for this vulnerable population. The absence of dedicated geriatric clinics further limited the capacity to address the unique health needs of older adults.

> *Many patients in our wards are elderly, dealing with various health issues beyond just hypertension. However, most staff, especially younger nurses, often aren't fully aware of the specific needs of geriatric patients. If we could provide proper orientation on the aging process, their developmental process, and how to respond to behavioral changes, it would greatly improve care. With this awareness, even the existing services could become much more effective and empathetic [HCP_04, Nursing In-charge, 43 years].*
> *There is a difference between other patients and elderly patients. It would be better if there was a spe-*

cialized clinic with experts running it. If there were a geriatric expert, it would be even better. Along with that, for each and every patient, like I have to see 50 patients, I can't counsel them all effectively." [HCP_10, DM cardiologist residence, 36 years].

High patient load and limited staffing

Healthcare providers noted an increase in patient load after insurance expansion, as more patients chose public facilities due to lower out of pocket costs. This rise in patient numbers strained staff capacity, resulting in heavier workloads and longer wait times, potentially affecting the quality of care.

"Because of insurance, the patient flow has increased. Staff who used to see 10 patients now see 15 or 16 regularly, which increases their workload." [HCP_09, Health Assistant, 42 years].

One of the persistent challenges highlighted by healthcare providers was the overwhelming number of patients relative to the limited availability of medical staff and time. This mismatch was especially problematic in outpatient settings, where only one or two doctors were often responsible for attending to a large number of patients, including elderly individuals who typically require more time, attention, and follow-up care. This not only compromised service quality but also led to delays, long waiting times, and reduced continuity of care.

Many patients, particularly older adults, expressed a desire to be seen by the same doctor during each visit. However, healthcare providers noted that this was difficult to fulfill due to the sheer volume of patients and staffing constraints. From the institutional perspective, consistently assigning the same doctor to individual patients was not feasible with the current human resource capacity.

Those patients always want to consult the same doctor, which is very difficult, right? And it's also because of the patient overload. From the hospital's perspective, it's not that easy either because it's not easy to provide that many staff. [HCP_10, DM cardiologist residence, 36 years]

One health worker explained that due to limited staff at Primary Health Center affected when the doctor was unavailable due to leave or other obligations. In such cases, patients were left without care, and healthcare providers expressed discomfort and helplessness in not being able to meet patient needs.

If some doctors take leave, it can be a bit difficult because there are people who really need to see a doctor at that time. When doctors aren't available, we also feel a bit uncomfortable since we can't provide the services they need. It becomes a challenge. If there were at least 2 or 3 doctors, then even if some took leave, at least one would still be available. [HCP_17, Senior ANM, 36 years]

System-level rigidity and inefficiencies

Health care providers explained how bureaucratic and procedural delays cause disruption in care delivery. Inflexible system rules such as strict medication refill schedules and administrative delays such as budget and procurement led to ineffective service delivery causing dissatisfaction among patients.

Why does it matter if I get my medicine on the 28th instead of the 31st? If I'm going on a 15-day religious tour and today is the 25th, how can I return just for my medicine on the 31st? I'm not misusing it—there's no harm in giving it a few days early. But here, even if someone comes a day early, they're told to come back the next day. Many spend 500 rupees on travel just to collect medicine worth 300, and it's really frustrating. [p3, patient, 70 years]
Sometimes the problem is not just the small budget, but delays in releasing it. Because of that, the process to buy medicines hasn't even started on time. We've had to buy medicines from our own budget for 2 or 3 months, but it's still not enough. [HCP_17, Senior ANM, 36 years]

Policy level challenges in health seeking

At the policy level, systemic and structural limitations in financial protection schemes and eligibility regulations contributed to economic insecurity and healthcare inaccessibility among hypertensive older adults. Participants highlighted that existing governmental policies were often inadequate, inconsistently implemented, or exclusionary.

Poor refilling system of medicine

Even when patients were enrolled in insurance, inconsistent availability and vague coverage of essential medications like antihypertensives limited the scheme's effectiveness. Participants experienced confusion, delays, and additional costs when trying to obtain medicines that were reportedly included in the benefit package.

The expenses are there, but what can we do? Insurance doesn't cover the pressure medicine. Sometimes they say something, and we have to wait in line, which gets frustrating and we have to go somewhere else. The cost problem is always there. [C5, caretaker, 35 years]

Sometimes they say it's available. Other times, they say that the medicines do not fall under the insurance category. [p1, patient, 77 years]

Exclusion for health insurance due to legal status

A lack of citizenship or legal documentation prevented some individuals, especially migrants or cross-border residents from enrolling in the national health insurance program. This legal exclusion translated into increased financial vulnerability and limited access to public health entitlements.

Not having citizenship has really affected me. Since I'm from India, I haven't been able to get Nepali citizenship. Because of that, I can't get insurance, and it's putting a lot of financial pressure on me. [p7, patient, 70 years]

Lack of structured policy programs for older adults

Lack of information on health seeking needs and opportunities for older adults. There are no community health care providers focused on providing health education to them.

During our community visits, we primarily focus on providing information to postpartum mothers, pregnant women, and children under five years of age. Older adults are not specifically sought out during these visits; we only share information with them if we happen to meet them while visiting the prioritized groups. [HCP_02, FCHV, 51 years]

Gap in policy implementation

Health coordinators and providers reported substantial discrepancies between government policies designed to support older adults and the reality of service delivery. Although official policies promise free or subsidized care, hidden costs and informal charges persist. This disconnect undermines seniors' access to essential care and reflects a lack of respect for their contributions and rights.

The reason senior citizens aren't accessing the services they need is because there's a big gap in how things are planned and what actually happens. When we look at policies made for senior citizens by the government, they don't match what is really needed. For example, hospitals might say they provide free services to senior citizens, but in reality, they still charge money in different ways. There isn't enough respect for the work and contributions these older adults have made in their lives. They deserve proper care as part of their basic health rights. The acts and policies sound good on paper, but it doesn't really work in practice which creates major problems in making sure senior citizens get the care they need. [HCP_11, Health coordinator, 50 years]

Health workers also pointed out their limited capacity to initiate improvements without government leadership and funding, emphasizing the need for timely medicine procurement and insurance program implementation to improve care continuity.

The government needs to take the initiative, but health centers can provide necessary services to patients as needed. We cannot start new initiatives on our own or bring funds independently. However, if the government or local bodies take the lead, timely tenders and regular supply of medicines can be ensured. If more medicines are available and insurance programs are implemented, things would become easier. [HCP_13, Medical officer, 33 years]
If geriatric services are provided in an integrated way through proper policy, then it would be possible to offer such seamless services. That way, many elderly citizens could be brought into care and we could make it easier for them. [HCP_14, Senior HA, 52 years]

Discussion

This study identified multi-level challenges that hinder older adult's access to health services. At the individual level, factors such as age-related physical and cognitive limitations, limited health autonomy, competing priorities, and perceived lack of necessity for care impeded health-seeking behaviors. Interpersonally, lack of family support, emotional neglect, and communication challenges further contributed to disengagement from care. At the community level, inadequate healthcare infrastructure, underutilization of local services, and transportation challenges made accessing care difficult. Organizational challenges included poor geriatric-focused infrastructure, lack of specialized training, fragmented service flow, long waiting times, and limited staffing. At the policy level, systemic issues such as exclusion from health insurance, lack of structured programs, inefficiencies in medicine refills, and gaps in policy implementation created significant challenges.

Individual level

At the individual level findings showed that age-related cognitive and functional decline including osteoarthritis, hearing loss and memory difficulties was a major challenge to seek care for hypertension, similar to the findings from previous research emphasizing its role in limiting chronic disease self-management in older adults [19, 20]. Participants also described a low perceived need

for routine hypertension care, particularly in the absence of symptoms, a pattern similarly observed in older populations in Bangladesh and India where silent chronic conditions like hypertension are often undervalued unless acute symptoms emerge [21, 22].

Similar to other studies, competing priorities set by older adults in their household work was a challenge for seeking care, with time constraints related to household responsibilities and caregiving roles as significant challenges for older adults seeking care [23]. Moreover, limited health autonomy, especially among older women, hindered timely access to services, a finding consistent with household survey data from Cambodia, the Philippines, and Vietnam, where women faced multiple sociocultural and structural challenges to healthcare access [24]. This gendered limitation in health decision-making is echoed globally, with older women in both LMICs and HICs reporting lower autonomy and higher vulnerability to unmet care needs, suggesting that gender-sensitive interventions are critical across contexts [25].

Interpersonal level

Interpersonal challenges further compounded these individual challenges. Participants frequently said inadequate family support, emotional neglect, and poor communication with health care providers. These reflect broader demographic and social transitions in South Asia, where traditional multigenerational households are giving way to nuclear families due to urbanization and labor migration. This shift has left many older adults without essential caregiving support, contributing to psychological distress and treatment discontinuation [26, 27]. Similar demographic transitions are also documented in Sub-Saharan Africa and Latin America, where urban migration and declining fertility have reduced traditional caregiving networks [28]. Even in high-income settings, smaller family sizes and geographic dispersion have produced comparable gaps, although state-supported social care partly mitigates these effects [29].

Additionally, similar to our study, unclear and inadequate communication with healthcare providers emerged as a critical challenge, which participants felt was mitigated to some extent by the involvement of supportive family members [10]. The communication gap between providers and older adults exacerbated by hearing loss and hesitation to seek clarification, which was also a recurring code, suggesting a need for more age- sensitive communication strategies.

Community level

At the community level, poor geographic access, inadequate transport, and a lack of trusted local healthcare services were reported. Despite the availability of basic health care centers at local level, these findings mirror challenges documented in rural Bangladesh, where long distances and travel costs deter care-seeking among older adults with chronic conditions [30]. Similar to this, a study reviewing primary care management of type II diabetes in West African countries (Nigeria, Ghana, Senegal) identifies health system deficiencies, costs, infrastructure gaps, and transport difficulties as significant barriers to chronic disease care utilization in rural Sub-Saharan Africa [31].

The underutilization of community-based services in our study may also reflect low trust in service quality consistent with other studies in Nepal that show patients often bypass local clinics for higher-level facilities despite added burden [11]. This has greater consequences in LMICs where bypassing translates into greater financial and logistical strain.

Organization level

From an organizational perspective, fragmented service delivery, insufficient staffing, long waiting times, and absence of geriatric-focused care infrastructures emerged as key challenges. These findings are aligned with global evidence from low- and middle-income countries (LMICs), where health systems remain underprepared to manage the long-term and complex care needs of aging populations [32]. However, in geriatric is integrated into primary care through developed innovative solutions such as nurse-led hypertension clinics in the UK, task-shifting to community health workers in Sub-Saharan Africa, and telehealth follow-ups in Australia and the US, that shows scalable models [33]. For Nepal, adopting such approaches could offer cost-effective alternatives to physician-centric, hospital-based models, particularly in addressing the growing burden of chronic conditions among older adults. Moreover, absence of trained geriatric providers and dedicated OPDs for older patients, despite national policy emphasis on age inclusive health care system. Similarly, long queues, poor provider communication, and lack of tailored services for older adults have been consistently cited as challenges to chronic disease management in other settings as well [34].

Policy level

At the policy level, systemic issues such as the lack of structured geriatric care programs, exclusion from health insurance schemes, and inefficiencies in the medicine refill process were evident. Despite Nepal's "Free Essential Health Services" policy, which mandates free antihypertensive medications at primary health centers, participants reported frequent stock-outs and were compelled to purchase medicines out-of-pocket, indicating a critical gap between policy and implementation [35]. This gap reflects a global pattern where in high income countries like the US, older adults still encounter affordability

issues due to medication co-payments under Medicare [36]. Furthermore, the absence of targeted geriatric policies, coupled with limited pension coverage and social welfare, has left many older adults reliant on family support or personal savings an issue noted in several LMICs [37].

Using socio-ecological models, this study provided multi-level challenges to seeking care. Through an iterative approach and code saturation, the study was able to capture diverse experiences and perceptions. The use of both IDIs and FGDs helped include a wide range of voices from older adults living with hypertension to caregivers and healthcare providers thus enriching the findings. These perspectives highlighted challenges at multiple levels, including personal, family, community, organizational, and policy-related challenges. By engaging various stakeholders, the study offers a multidimensional understanding of the challenges the older adults face while seeking hypertension care.

Our findings extend a growing body of research from Nepal showing that older adults face multiple, overlapping barriers to care. Several qualitative and mixed-methods studies report that affordability and poor geographic access, long travel distances and unreliable transport regularly prevent older people from reaching services, especially in rural areas [10]. Other studies have shown that the cost of medicines and weak socioeconomic protections (including limitations of the old-age allowance), which leave many older adults dependent on family support or unable to afford ongoing chronic-disease treatment [38]. Researchers have repeatedly noted the importance of social support and the lack of age-friendly services: when family or escorts are unavailable, older people often skip appointments; and clinics rarely provide geriatric-sensitive care or clear pathways for elderly patients, undermining acceptability and continuity of care [10]. Together, these studies and our own showing that the barriers are structural (transport, cost, weak policy implementation), social (limited caregiver support, cultural norms), and disease-specific (medication refills, asymptomatic low perceived need), which combine to produce persistently low healthcare utilisation among older adults with chronic conditions like hypertension in Nepal.

The findings from this study shows barriers across different levels of the socio-ecological model interacting and reinforcing one another, creating a compounded effect on care-seeking among older adults. Individual-level limitations such as reduced mobility and low health autonomy are intensified by interpersonal gaps like limited family support, which in turn diminish an older adult's ability or motivation to seek care. These personal and relational constraints are further magnified by community barriers such as poor transportation and lack of trusted local services that physically and socially isolate older adults from healthcare access. At the organizational level, long waiting times and the absence of age-sensitive communication approaches discourage older adults from visiting facilities, reinforcing perceptions of poor service quality at the community level. Finally, weak policy implementation evident in inconsistent medicine supply and exclusion from insurance schemes undermines both the capacity of health organizations and the trust of individuals and communities. These interlinkages highlight that addressing one level in isolation may yield limited results; rather, interventions must adopt a holistic approach that simultaneously strengthens individual empowerment, family engagement, community trust, organizational readiness, and policy enforcement. Such an integrated perspective aligns with the core principles of the socio-ecological model, emphasizing that health behaviors and access are shaped through dynamic interactions across multiple, interconnected layers of influence.

Despite these strengths, the study is not without limitations. As is often the case in qualitative research, participant narratives may have been influenced by social desirability bias or personal discomfort, leading to potential under-reporting or over-reporting of experiences. Furthermore, the possibility of selection bias cannot be ruled out; those who agreed to participate may differ in meaningful ways from those who declined or were unavailable, potentially limiting the generalizability of the findings to the wider population of older adults.

Implication of the study

The study's findings have important policy and programmatic implications. They can guide the development of geriatric-friendly hypertension services and strengthen family and community support systems for older adults. This study also highlights the need to close the policy-to-practice gap by ensuring reliable medicine supply and expanding health insurance coverage to include chronic disease management. To build on these insights, future quantitative research with larger representative samples is recommended to assess the scale of these challenges and evaluate potential interventions.

Conclusion

This study explores the complex, multi-level challenges faced by older adults with hypertension in Nepal, extending from individual challenges such as limited autonomy, competing priorities, and age-related physical or cognitive decline, to interpersonal challenges including weak caregiver involvement, neglect, and communication gaps. At the community level, underutilization of local services, inadequate community-based care, and transportation challenges further hinder access, while organizational challenges such as poor geriatric-centered

infrastructure, lack of specialized training, fragmented service flows, and high patient loads compromising the quality of care. These challenges are aggravated at the policy level by the exclusion of geriatric needs from health insurance, weak program structures, and gaps in implementation. To address these interconnected challenges, responsive measures are needed: programs that strengthen community-based hypertension monitoring and actively involve caregivers; organizational reforms that invest in geriatric-friendly infrastructure and health care worker training; policy interventions that expand insurance coverage and ensure local-level enforcement; and research that pilots and evaluates community- and caregiver-led models of care. Taken together, these targeted actions are critical for building an equitable, accessible, and age-inclusive health system for managing hypertension among older adults in Nepal.

Supplementary Information

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Supplementary Material 1.

Supplementary Material 2.

Supplementary Material 3.

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Author contributions
PS, BK, and VV were responsible for conceptualization and design of the study. PS and YA was responsible for obtaining ethical approval. PS, YA and IP were responsible for data collection. PS, YA and IP were responsible for data analysis and interpretation. PS, BK, VV, YA and IP were responsible for drafting the manuscript. PS and YA were responsible for editing and revising the manuscript. All authors have read and approved the manuscript. PS is responsible for the overall content as the guarantor.

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Data availability
The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate
We obtained ethical approval from the Ethical Review Board of the Nepal Health Research Council (IRC registration number: 113_2024/Ref no: 2132) and from the Kathmandu University Institutional Review Committee (KU-IRC: 165/24). We also obtained permission from the Dhulikhel municipality before beginning data collection. During the consent process, we explained the potential risks and benefits of participating in the interviews to the participants. All respondents provided written informed consent. Literate participants signed the consent form, while illiterate participants gave their thumbprint along with the signature of a caretaker or witness. We informed participants that their involvement was voluntary, that they could skip any questions they did not wish to answer, withdraw at any point during the interview, and that their responses would remain confidential. The study obeyed the ethical standards defined in the Declaration of Helsinki.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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